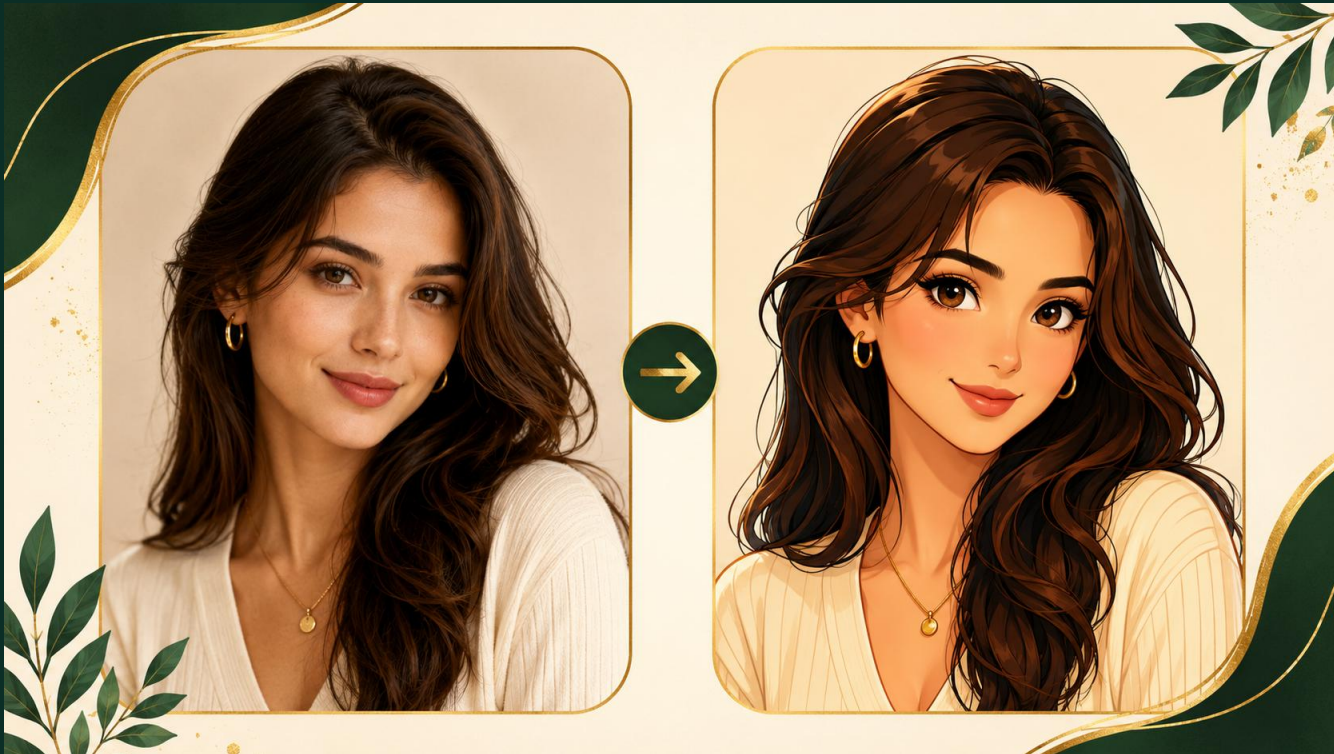


Traditional Image Processing Cartoon Style App

Java OOP Challenge Project in VS Code



Challenge Overview

- Pure traditional image processing
- No AI or deep learning
- Object-oriented Java architecture
- Real-time cartoon filter preview
- Runs directly in VS Code
- Student challenge style project



Object-Oriented Java Architecture

ImageLoader

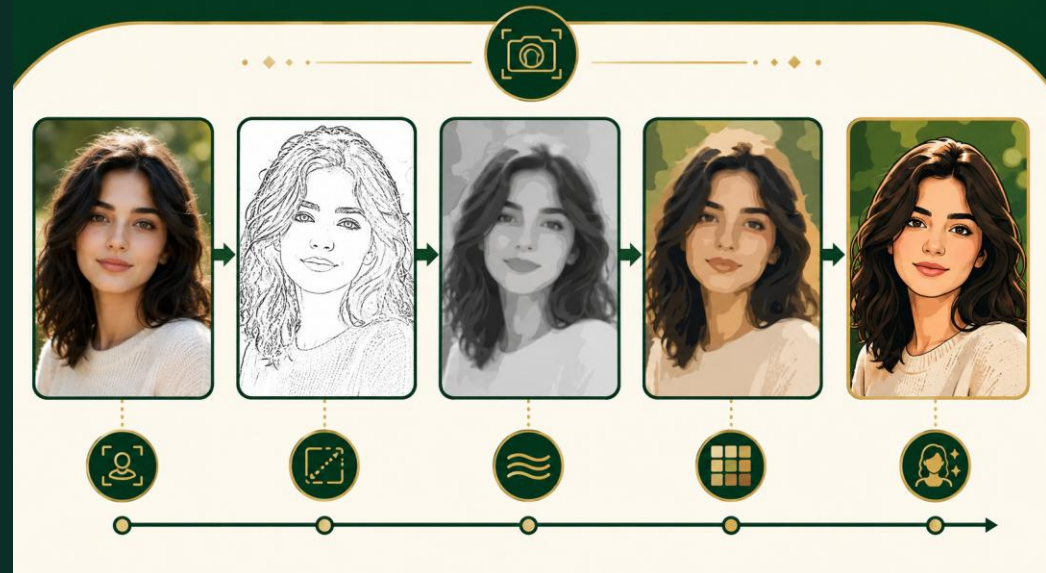
GrayConverter

EdgeDetector

ColorQuantizer

CartoonRenderer

```
BufferedImage img = ImageIO.read(file);  
GrayImage gray = converter.toGray(img);  
EdgeMap edge = detector.detect(gray);  
CartoonImage out = renderer.render(edge);  
display.show(out);
```



Traditional Cartoon Processing Pipeline



Core Features

- ✓ Gaussian blur smoothing
- ✓ Sobel edge extraction
- ✓ Color quantization
- ✓ Adaptive thresholding
- ✓ Real-time preview window
- ✓ File save/export support
- ✓ Clean Java Swing GUI
- ✓ VS Code compatible



Build Your Own Cartoon Filter App

Traditional image processing is fast, explainable, and fun to build.
Perfect for Java OOP practice and programming challenges.

